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Effect of Heat Flux on Bubble Coalescence Phenomena and Sound Signatures During Pool Boiling

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Boiling heat transfer is extensively used in various industrial applications to efficiently dissipate a large amount of heat by maintaining low surface to fluid temperature differences. The maximum heat flux dissipated during boiling is limited by the critical heat flux (CHF). Difficulties in visualizing the boiling process and monitoring surface temperature make it difficult to identify the impending CHF condition. As a result, larger factors of safety are employed, and the equipment is operated at considerably lower heat fluxes. This study focuses on identifying acoustic signatures of different nucleate boiling regimes. The bubble nucleation and coalescence along with bubble collapse lead to variation in acoustic emission patterns during boiling. In this work, acoustic signatures representative of the impending CHF condition are identified during pool boiling of water over plain and enhanced copper substrates. It was observed that sound was dominant in two frequency regions (400–500 Hz dominant throughout the nucleate boiling, and 100–200 Hz dominant at heat fluxes > 100 W/cm^2). However, just before CHF, a sudden drop in amplitude was observed in the high frequency region (400–500 Hz), while the amplitude in low frequency region (100–200 Hz) continued to rise. This combination of the amplitude trend has potential to predict the approaching CHF condition. This is the first study that correlates high-speed images with the acoustic trends at different stages of nucleate boiling and compares experimental frequency with Minnaert frequency for coalescing bubbles. [DOI: 10.1115/1.4050088]

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Introduction

Increasing demand of technology has led to an evolution of innovative approaches, solutions, and ideas for providing effective cooling solutions at high heat fluxes. While the demand has led to a significant increase in the use of electronic devices, the cost to cool them continues to rise. Various single-phase cooling approaches have been used to remove the heat generated on these devices [1]. Owing to efficient heat removal with large amount of latent heat, pool boiling offers an attractive passive cooling technique. However, it is important to recognize the limitations of this cooling technology. During pool boiling, vapor bubbles are generated that effectively carry away the heat and maintain the heater surface temperature within design limits. However, heat dissipation capacity of the heater surface is limited by critical heat flux (CHF), a condition at which an insulating vapor layer covers the heater surface due to excessive bubble generation rate. Researchers have successfully pushed the CHF limits to higher values by implementing various enhancement techniques including modified surface geometries and shapes, formation of separate liquid–vapor pathways [2], and formation of microporous coatings [3–5].

For safe operation of the devices, detection of the impending CHF condition is one of the most critical factors. The CHF is initiated within a very short time duration of the order of milliseconds, and if corrective action is not taken, such as reducing the heat flux or providing secondary coolant stream, surface temperature rises to very high values that can cause catastrophic damage to the system. Additionally, monitoring of the CHF can be very difficult for systems without continuous temperature monitoring. Thus, the detection of CHF condition is a critical consideration in designing a pool boiling system. In industries such as fuel-fired or nuclear power plants where industrial boilers play an integral role and failure of such systems can be catastrophic, the noninvasive acoustic monitoring can be used as additional tool with temperature method to safeguard these systems. In data center applications, where immersion cooling methods for cooling high power and high heat generating CPU chips are implemented, failure of such systems can be very expensive and can lead to loss of valuable data. Acoustic monitoring method can be used at such sites.

Current systems for identifying the CHF are mostly dependent on temperature data. However, they are not very reliable due to the random and rapid nature of conditions initiating CHF. In the absence of such temperature monitoring, larger safety factors are employed by lowering the operating heat fluxes. Acoustic monitoring tools can provide an additional safety feature and assist in increasing the working range and efficiencies of the boiling equipment.

Literature Review

Some of the important research work related to the basics of sound generation for air and vapor bubbles is first presented in this section. Minnaert [6] was the first researcher who laid the basic foundation on understanding the dynamic behavior of air bubbles in a liquid. Minnaert expression showed that oscillating frequency and bubble radius are inversely proportional, which was also confirmed by experimental observations. Manasseh and Risso [7] observed a rise in sound pressure initially during air bubble coalescence. Their results suggested that volume oscillation during coalescence initiates a pressure equalizing mechanism in coalescing air bubbles that is the main driver for sound emission. Kracht and Finch [8] experimentally showed that a coalesced air

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bubble has a higher amplitude of oscillation compared to the individual coalescing bubbles.

The bubble generation process in pool boiling is very rapid, and hence, coalescence becomes a dominating factor in sound generation. In addition to coalescence, the sound is generated due to bubble growth, oscillations during departure, and bubble collapse [9]. According to Osborne and Holland [10], the sound frequency distribution is dependent on the material and size of the boiling surface. For smaller wires, they observed that as the wire diameter decreases, the number of peaks in the frequency domain increases, but for larger wires, the frequency of the peaks increases with heat input.

Sinha et al. [11] observed a gradual increase in amplitude without any significant changes in the frequency domain (200–250 Hz) in the nucleate boiling region using a CHF detection feedback system. A sudden rise in amplitude was observed at CHF followed by complete sound attenuation in film boiling. They noted a sudden shift in the frequency region (400–500 Hz) after reaching the CHF and attributed it to the change from mushroom shape bubbles before the CHF to smaller individual bubbles post-CHF. Aoki and Welty [12] studied the sound generated during saturated boiling on a horizontal copper disk using pentane as the working fluid. They observed that boiling sound increased with heat flux, but a drop in sound level was observed near CHF due to formation of a vapor layer. They noted dominant frequencies to be in the 500–1000 Hz range during the boiling process. Westwater et al. [13], while working with methyl alcohol, observed a continuous rise in sound pressure level in nucleate and transition boiling regions with increasing heat flux. Schwartz and Siler [14] used a hydrophone to capture the sound while minimizing the background noise during boiling. In their study, they found that the sound amplitude increases with an increase in heat flux in the nucleate boiling region and then drops suddenly at film boiling transition after CHF. They noted the sound to be mainly in the low frequency domain from 25 to 700 Hz. Mohanty and Singh [15] monitored the boiling sound by using two microphones in a closed system for continuous mapping of sharp resonant peaks in frequency domain to detect the varying liquid level in the system. Frequency domain during experiments was observed to be below 1000 Hz with multiple peaks around 250 and 700 Hz. Tang et al. [16] studied boiling on a small-sized heater and observed a decrease in peak frequency with increasing bubble size. Initially, amplitude peaks were observed around 5 kHz as bubble size was reduced due to subcooling. However, an increase in wall superheat and liquid temperature shifted the peak toward the low frequency region. Dominant peaks were observed around 100–200 Hz for fully grown coalesced vapor bubbles.

Several studies [12,17] have focused on studying the acoustic signatures during boiling, but they have not focused on characterizing the region close CHF and identifying the impending condition. Currently, there are no acoustic criteria available for detection of impending CHF. With new improved tools available for acoustic sensor data analysis, a new approach is developed in this study to characterize acoustic emissions during pool boiling.

Hypothesis

During pool boiling, the vapor bubbles nucleate, grow, and depart from a heater surface. The bubble coalescence, departure from the heater surface, and bursting at the liquid surface lead to energy release in the form of sound. It is hypothesized that acoustic signatures are associated with the bubble coalescence and departure states that are unique at different heat fluxes. Distinctive boiling sound changes occur at various heat fluxes that are indicative of impending CHF resulting from initiation of vapor blanketing on the heater surface. Acoustic mapping can thus be used to detect the state of a boiling system.

Surface Preparation

A plain copper heating surface was made from copper alloy 101 using computer numerical control (CNC) technique, and the surface was smoothened using a grinding/polishing wheels. The average roughness (R_a) of the boiling surface was ~1.6 μm which was measured using a laser confocal microscope (see Fig. 1(*a*)). The microporous surface was prepared using a sintering technique on the plain copper surface using 5% graphene nanoplatelets-copper particle mixture. A similar technique as used by Rishi et al. [18] was implemented, and the morphological structure of the porous coating was characterized using a laser confocal microscope (see Fig. 1(*b*)).

Experimental Setup

Figure 1(*c*) shows a schematic of the pool boiling test setup. It consists of a water bath formed by a 14 mm × 14 mm × 38 mm quartz glass enclosure, a copper test chip, and a heater block both made of copper alloy 101. The test chip is placed in a ceramic chip holder with aligned holes on both the block and the chip to facilitate insertion of three thermocouples for heat flux measurement. A rubber gasket is used to seal the contacting surfaces with the enclosure. Four stainless steel screws hold the middle garolite plate with the bottom aluminum plate. A water reservoir is secured between this middle garolite plate and top aluminum plate. An auxiliary cartridge heater (60-VDC, 200 W) is fitted on the aluminum plate, and a small hole is provided in the top plate to insert a thermocouple to measure the water bath temperature.

The bottom section consists of a copper heater block with four 120-VDC, 200 W capacity cartridge heaters. A Grafoil (carbon fiber) sheet is placed between the heater and the test surface to minimize heat losses. The heater block rests on a ceramic block placed on an aluminum plate supported by four compression springs to provide the required degree-of-freedom to account for thermal expansion and maintain good thermal contact between the test chip and the heater block [19].

Thermal Data Acquisition. A National Instruments data acquisition system cDAQ-9172 with NI-9211 module was used to record the temperature data from four thermocouples as shown in Fig. 2(*a*). Three thermocouples were inserted in the stem of the test chip, and one thermocouple was inserted from the top cover to measure the bath temperature. A LabVIEW VI was designed to display the temperature data from thermocouples and calculate the heat flux. Each test chip has surface area of 17 mm × 17 mm, with the central 10 mm × 10 mm area exposed to the boiling liquid and the remaining region covered with a Kapton tape to act as an insulation. The three thermocouple holes in the test chip are 3 mm (Δx) apart, and the distance of the top thermocouple hole from the test surface is 1.5 mm (x_1).

Heat Flux is calculated using steady-state 1D heat conduction equation

$$q'' = -k_{Cu}\frac{dT}{dx} \tag{1}$$

where the temperature gradient dT/dx is calculated using the three-point backward Taylor's series approximation

$$\frac{dT}{dx} = \frac{3T_1 - 4T_2 + T_3}{2\Delta x} \tag{2}$$

T_1, T_2, and T_3 are the temperatures corresponding to the top, middle and bottom thermocouples, respectively. The test chip surface temperature T_{wall} is obtained using Eq. (3) with the distance x_1 between the surface and top thermocouple, which is equal to 1.5 mm

$$T_{wall} = T_1 - q''\left(\frac{x_1}{k_{Cu}}\right) \tag{3}$$

Acoustic Data Acquisition/Microphone Setup. Two Ecopro EO-200 condenser microphones are used for sound measurement as shown in Fig. 1. Their positions are kept fixed during the tests.

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The bottom microphone is placed 1 cm away from the test surface, while the top microphone is placed 0.5 cm from the top free surface of water bath to ensure that there is no contact between the microphone and the experimental setup. These microphones are unidirectional cardioid microphones having sensitivity in 50–10,000 Hz frequency range. It was observed that the distance from the water bath influenced only the amplitude while the sound pattern remained unaltered.

Water level in the reservoir is maintained 3 cm above the middle garolite plate throughout the experiments. The microphones are kept at two fixed locations for all the tests, and they are connected to Steinberg UR 44 audio interface as shown in Fig. 2(*b*). An audio interface is connected to the computer using a USB input, and sound is recorded in Audacity software at a sample rate of 44,100 Hz. Since the sound recorded by both bottom and top microphones were in the same frequency domain, the data from the top microphones was considered for all the tests as its sound output was higher due to its proximity to the water bath. A preliminary evaluation of the water level indicated that the sound signatures were unaltered while the pool height was varied from 2 cm to 6 cm.

Uncertainty Analysis

Uncertainty in Boiling Data. During the experiments, precision and bias errors are present in the measurements. Bias errors arise from the calibration while precision errors occur because of sensitivity of the testing instruments. For this study, uncertainty analysis was performed as described by Patil and Kandlikar [20] for a similar setup. Each thermocouple was calibrated in the temperature range higher than the testing temperatures. Based on the factors such as the spacing between thermocouples, calibration data of thermocouples, and thermal conductivity of copper, total uncertainty in heat flux and heat transfer coefficient was calculated. Table 1 shows the bias uncertainty for different parameters under considerations. Uncertainty at higher heat fluxes (above 70 W/cm^2) is observed to be less than 5% for this study. This is in the same range as reported in previous studies [20,21].

Uncertainty in Acoustic Data. Precision uncertainty for boiling sound was calculated using a relative method as there is no true value for amplitude data. In this method, a soundtrack of 15 s is divided into three subsections of 5 s each. Amplitude data is converted from the time domain to the frequency domain for each subsection, and the frequency domain plots are smoothened using a moving weighted average. Now, each frequency will have three amplitude values, and the standard deviation of these amplitude values from mean value is calculated from Eq. (4), where the value of N is 3. The variation in amplitude with respect to frequency for different heat fluxes is shown in Fig. 3. Equations (4)

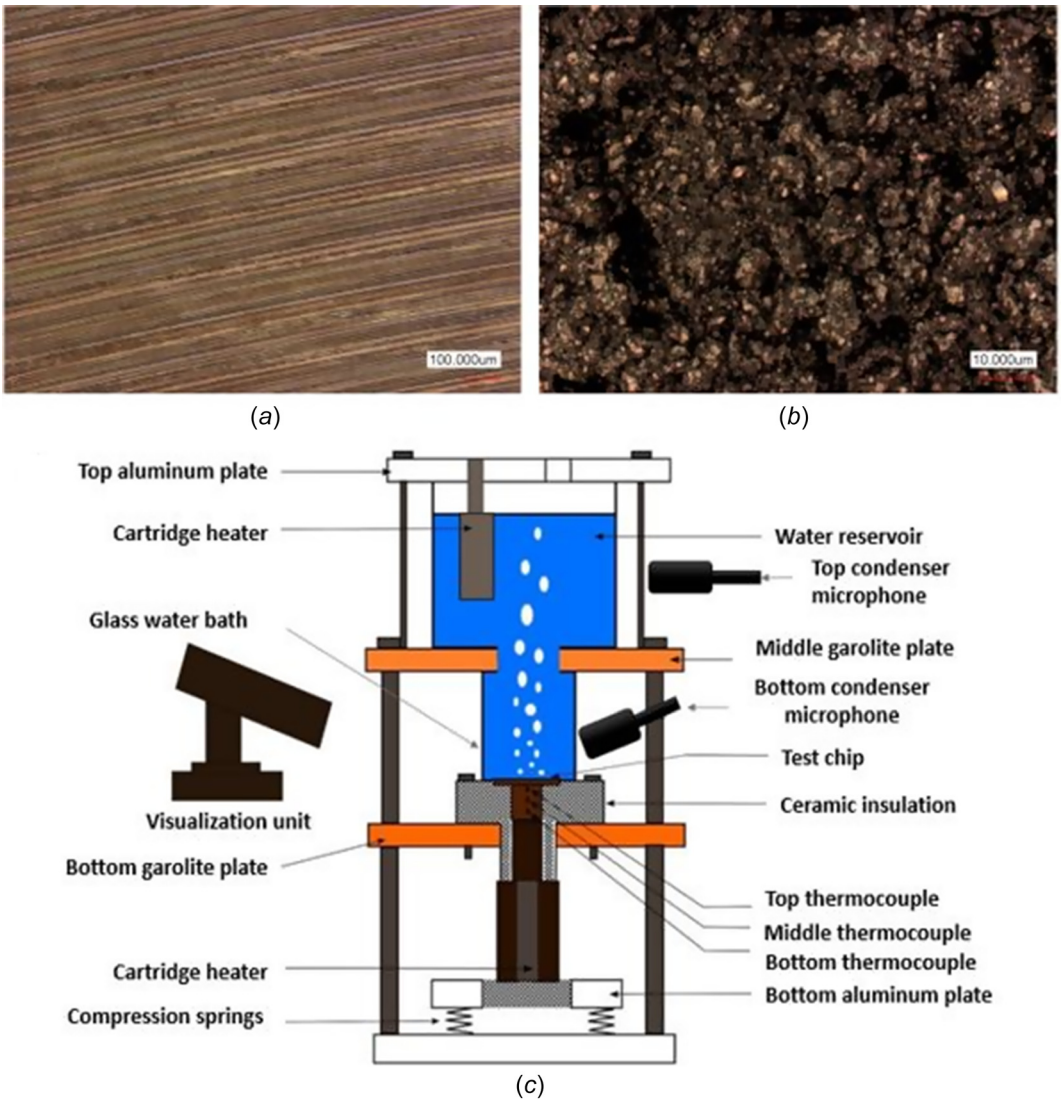


Fig. 1 Laser confocal microscope images showing (*a*) plain copper chip morphology, (*b*) microporous chip morphology, and (*c*) pool boiling test setup schematic

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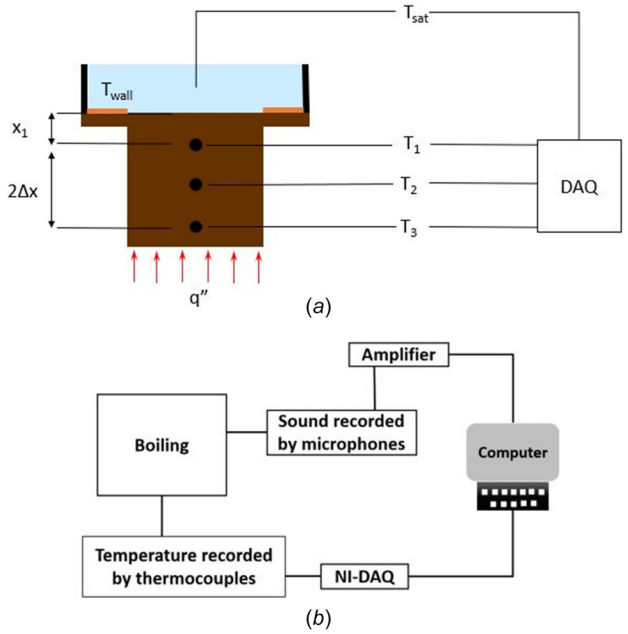


Fig. 2 (*a*) Schematic of data acquisition from test chip and (*b*) schematic of acoustic data acquisition during pool boiling

Table 1 Biased uncertainty for various parameters

Parameters	Value	Units	Biased uncertainty (U_P)	% Uncertainty
k_{Cu}	391	W/m-°C	9.00	2%
Δx	3.00×10^{-3}	m	0.001	3%
x_1	$1.70 \times 10^{-}3$	m	0.001	6%
T_T	Varies	°C	0.16	Varies
T_M	Varies	°C	0.40	Varies
T_B	Varies	°C	0.25	Varies

and (5) are used to determine the uncertainty at specific frequencies. The maximum uncertainty was observed at 450 Hz for 116 W/cm^2 and was less than 8% with respect to the mean amplitude value

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(a_i - \bar{a})^2}{N}} \tag{4}$$

$$U_s = 2*\sigma \tag{5}$$

Uncertainty in the microphone data was calculated by generating sound of desired frequency from Sonic application developed by Von Bruno software and recorded using the sensor used in this study. This sound was recorded in Audacity software, which was also used for recording the boiling sound. For each frequency, five tracks of 10 s each were recorded. Frequency domain plots for each track were generated in MATLAB. The generated frequency is considered as the true frequency value. The desired frequency was generated in the range of 150–600 Hz based on the frequency range recordings obtained during the pool boiling study. The exact frequency values generated were 150.0 Hz, 250.0 Hz, 403.0 Hz, 508.0 Hz, and 603.0 Hz. The frequency response for 403 Hz sample is shown in Fig. 3(*e*). Similar plots were obtained for four additional samples at each frequency. The dominant peaks for each sample show the recorded frequency values for the respective true frequency values. The standard deviation for each frequency was calculated using the five recorded tracks and the corresponding percentage error was evaluated as twice the standard deviation. The maximum error was 3.97% at 150 Hz frequency. Errors at other frequencies were below 0.5%. Since the frequency plots used in the boiling studies are in the range 0–600 Hz, the uncertainty bars for frequency are smaller than the symbol sizes and hence they are not shown in the boiling plots.

Results

Pool boiling tests were performed on a plain copper surface and a microporous surface. The recorded sound data were analyzed using a MATLAB code in time and frequency domians using fast Fourier transform (FFT). The plain copper surface attained a CHF of 130 W/cm^2, while microporous surface achieved a CHF of 204 W/cm^2. Since the acoustic emission detection method is highly sensitive, it captures the background noise and other signals as well in addition to the boiling sound. The background noise was canceled in Audacity software with inbuilt noise cancelation tool. Due to weak acoustic signals at lower heat fluxes, it is very difficult to analyze the sound data below 30 W/cm^2 since the sample size is small, and it becomes challenging to distinguish between the boiling signals from the noise. Acoustic emission signals from boiling become dominant above a heat fluxes of around 30 W/cm^2.

Amplitude Variation With Time. After extracting amplitude-time data, a 20 ms soundtrack is considered for analyzing the variations in pulse length and amplitude. Four different heat fluxes as shown in Fig. 4 were considered for the analyzing the plain copper surface. On analyzing a small section of the signal in a short time frame, it is clear that pulse length increases continuously with increasing heat fluxes. Also, time interval between the first pulse and the following pulse decreases with increase in heat flux.

With increase in heat flux, amplitudes over the entire frequency range continue to rise as seen in Fig. 4(*a*)–4(*c*). However, just before CHF, Fig. 4(*d*), the amplitudes are reduced. As the CHF condition approaches, lateral bubble coalescence increases, and vapor layer starts covering the heater surface resulting in a reduction in amplitude of the signal. Similar phenomenon of reduction in amplitude just before CHF was also observed for the microporous surface. Further details on the bubble coalescence patterns are discussed in Discussion section under high-speed imaging.

Dominant Frequency Domain of Boiling Sound. To understand the dominant frequency region at various stages during pool boiling, the sound amplitude intensity versus frequency plots were generated after converting the amplitude-time plot using FFT.

Plain Copper Surface. Figures 5(*a*)–5(*d*) show amplitude versus frequency plots at four different heat fluxes for a plain copper surface. It was observed that for both plain copper and microporous surfaces, the sound was detected mainly in the frequency region between 0 and 600 Hz, and no amplitude peaks were noted in the frequency range beyond 600 Hz till 10,000 Hz. These plots are used in comparing the bubble behavior with corresponding sound pressure levels and frequency distribution at different heat fluxes. As seen from Fig. 5(*a*), at lower heat fluxes (36 W/cm^2), peaks start to appear dominantly in the 400–500 Hz frequency range. As heat flux is increased from 36 to 71 W/cm^2, there is a gradual rise in sound amplitude intensity as observed in Fig. 5(*b*). It is also observed that an increase in heat flux only leads to increment in sound amplitude without significant change in the frequency domain. A similar variation of amplitude intensity with increase in heat flux has also been observed by other researchers [9,13,14].

With further increase in heat flux (Fig. 5(*c*) at 116 W/cm^2), in addition to peaks in 400–500 Hz, sound amplitude variations start to appear in lower frequency domain (100–200 Hz) as well. This trend was observed at higher heat fluxes (>∼100 W/cm^2). At 127 W/cm^2, when the heat flux is very close to the CHF condition,

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peak intensity in the higher frequency region (400–500 Hz) drops suddenly. However, peak intensity in the lower frequency region (100–200 Hz) still continues to rise (Fig. 5(*d*)). This is indicative of impending CHF condition.

Microporous Surface. Similar to a plain copper surface, sound amplitude intensity versus frequency plots were generated for microporous surface. Similar trend of increment and reduction of amplitude with heat flux was observed for the microporous surface as well. All dominant peaks were observed in the frequency range of 0–600 Hz. At lower heat fluxes (around 40–68 W/cm^2), amplitude peaks were observed in the frequency range 300–400 Hz, Fig. 6(*a*). With further increase in heat flux, these amplitude peaks start getting concentrated around 400–500 Hz range around 86–112 W/cm^2 as shown in Fig. 6(*b*). At heat fluxes above ~135 W/cm^2, the amplitude peaks in the low frequency region start appearing and their intensity increases with increment in heat flux. Figure 6(*c*) shows the dominant peaks in 100–200 Hz range and a continuous increment in amplitude peaks in 400–500 Hz frequency range. At very high heat fluxes, ~200 W/cm^2, Fig. 6(*d*), just before CHF, a sudden drop in amplitude is observed in the high frequency region (400–500 Hz). However,

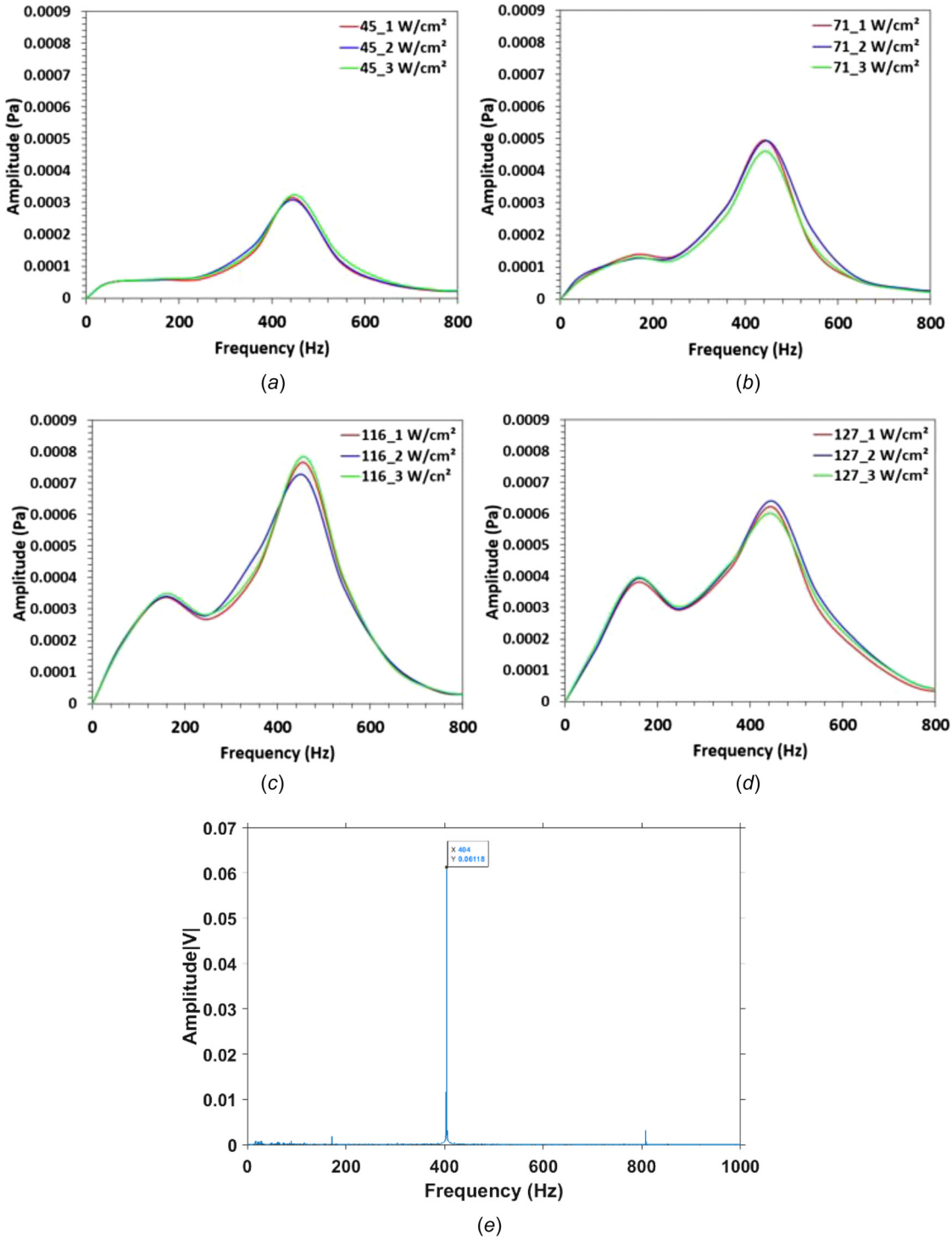


Fig. 3 (*a*)–(*d*) Uncertainty in amplitude at the same heat flux value, and (*e*) Peak output for 403 Hz frequency

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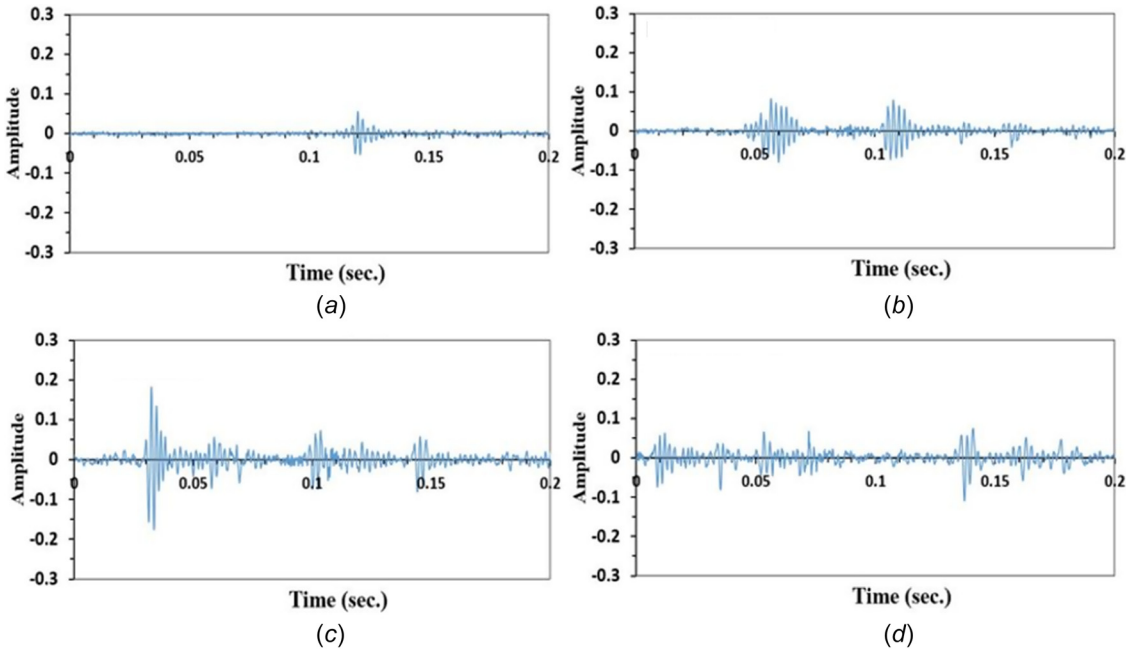


Fig. 4 Amplitude of signals in 20 milliseconds at different heat fluxes for a plain copper chip

peaks in the lower frequency region (100–200 Hz) continue to rise. The same behavior was noted for a plain surface indicative of imminent CHF condition.

This combination of a sudden drop of amplitude in 400–500 Hz frequency region and continued increment of amplitude in 100–200 Hz frequency region observed for both plain and microporous surfaces is seen as an indication of approaching CHF for the plain copper and enhanced surfaces.

Comparison of Acoustic Signature Trends of Plain Copper Surface and Microporous Surface. To compare the acoustic signature trends of a plain copper surface and a microporous surface, the amplitude–frequency curves shown in Fig. 7 were averaged and smoothened using moving weighted average method [22]. These curves provide a better understanding of the dominant frequency regions during nucleate boiling and variations in amplitude as a function of heat flux. Figure 7(*a*) shows the comparison

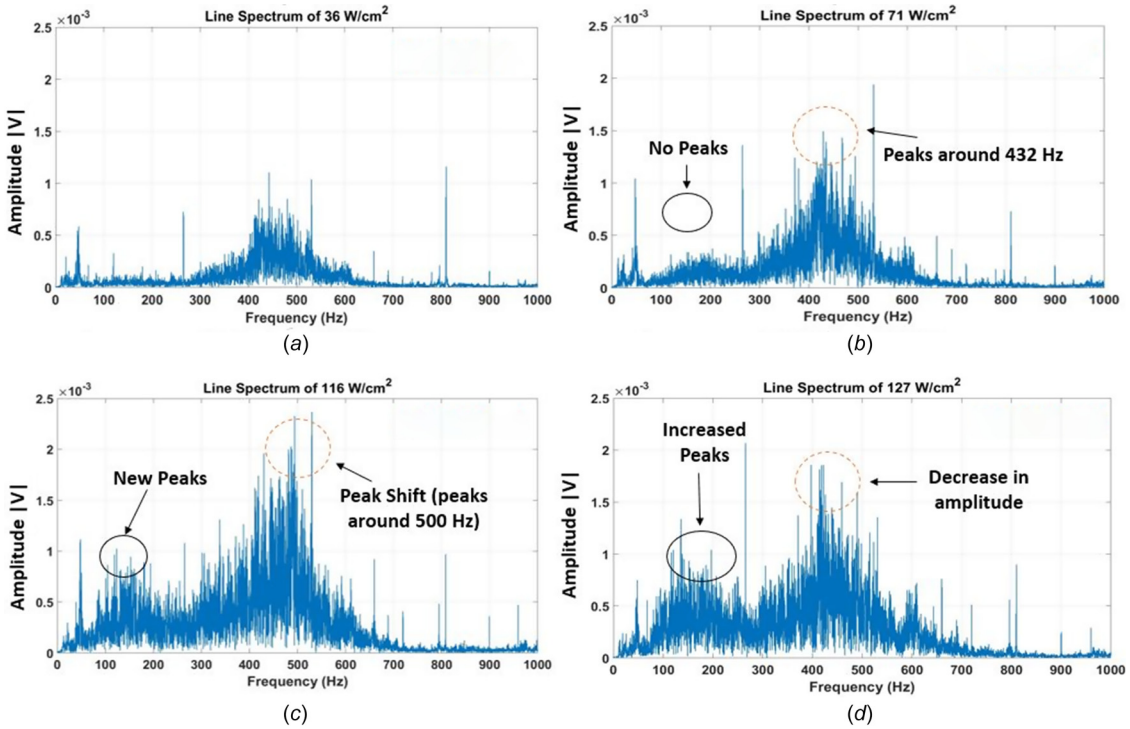


Fig. 5 Amplitude variation in frequency domain for a plain copper chip at varying heat fluxes (CHF = 130 W/cm^2)

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of average amplitude versus frequency plots for a plain copper surface and a microporous surface. To avoid clustering of data, the amplitudes for only higher heat fluxes (above ~100 W/cm^2) are shown.

Figure 7 shows a microporous surface also exhibiting a similar acoustic trend as that of the plain copper surface. The enlarged section in Fig. 7(*c*) shows maximum amplitude values at ~450 Hz for different heat fluxes with their respective error bars. From Figs. 7(*b*) and 7(*c*), it is clearly seen that for both plain copper and microporous surfaces, amplitude suddenly drops just before CHF in the high frequency region, while continues to rise in the low frequency region. From the enlarged sections, it is observed that these amplitude variations are larger than the respective error bars. This combination of sudden drop of amplitude in 400–500 Hz region and continued increment of amplitude in 100–200 Hz region right before CHF confirms that a continuous acoustic mapping can be used as a tool to predict the approaching CHF condition.

Discussions

High-Speed Visualization Study for Dominant Frequency Regions. Regions of interest for this study are identified as being in the frequency ranges of 400–500 Hz and 100–200 Hz. Sound peaks in 400–500 Hz frequency range start to appear as soon as the bubble coalescence starts. Sound in this high frequency region is seen to be due to collapse of fully grown coalesced vapor bubbles at the surface. As heat flux increases and boiling becomes more rapid, sound amplitude in the high frequency region increases. However, just before CHF, amplitude suddenly drops. It is hypothesized that this sudden drop of amplitude is due to the initiation of a vapor layer on the heater surface. Figure 8 shows the high-speed image of a coalesced vapor bubble captured at 71 W/cm^2 responsible for producing the sound in high frequency region which is further validated using Minnaert's frequency equation, Eq. (6), as described in the Comparison section.

At higher heat fluxes (>100 W/cm^2), peaks begin to appear in the low frequency region in addition to the high frequency region. Here, rapid bubble coalescence occurs and vapor lumps start merging vertically, forming vapor columns. These columns collapse at the liquid bath surface and generate sound which is dominant in the low frequency region. The time interval between the formations of these vapor columns is rather large, and hence, the sound produced by them is dominant in the lower frequency region. Thus, at higher heat fluxes, sound generation is due to combination of fully grown coalesced vapor bubbles (400–500 Hz) and vapor jet columns (100–200 Hz). High-speed images captured at high heat fluxes in Fig. 8(*b*) show the corresponding fully grown coalesced vapor bubble and vapor jet column responsible for the sound generation.

Comparison of Experimental Frequency and Minnaert Frequency for Boiling Bubbles. The experimental frequency ranges recorded for a plain copper surface were compared with the theoretical Minnaert frequencies. Minnaert [6] derived the relationship (Eq. (6)) between resonant frequency of an oscillating bubble and its radius by comparing the potential energy which bubble attains at minimum volume during oscillation to the kinetic energy at the equilibrium position. We captured high-speed images at different heat fluxes to measure the bubble diameters and calculated the Minnaert frequencies. Equivalent diameter of the vapor bubble was calculated by measuring horizontal and vertical diameter and

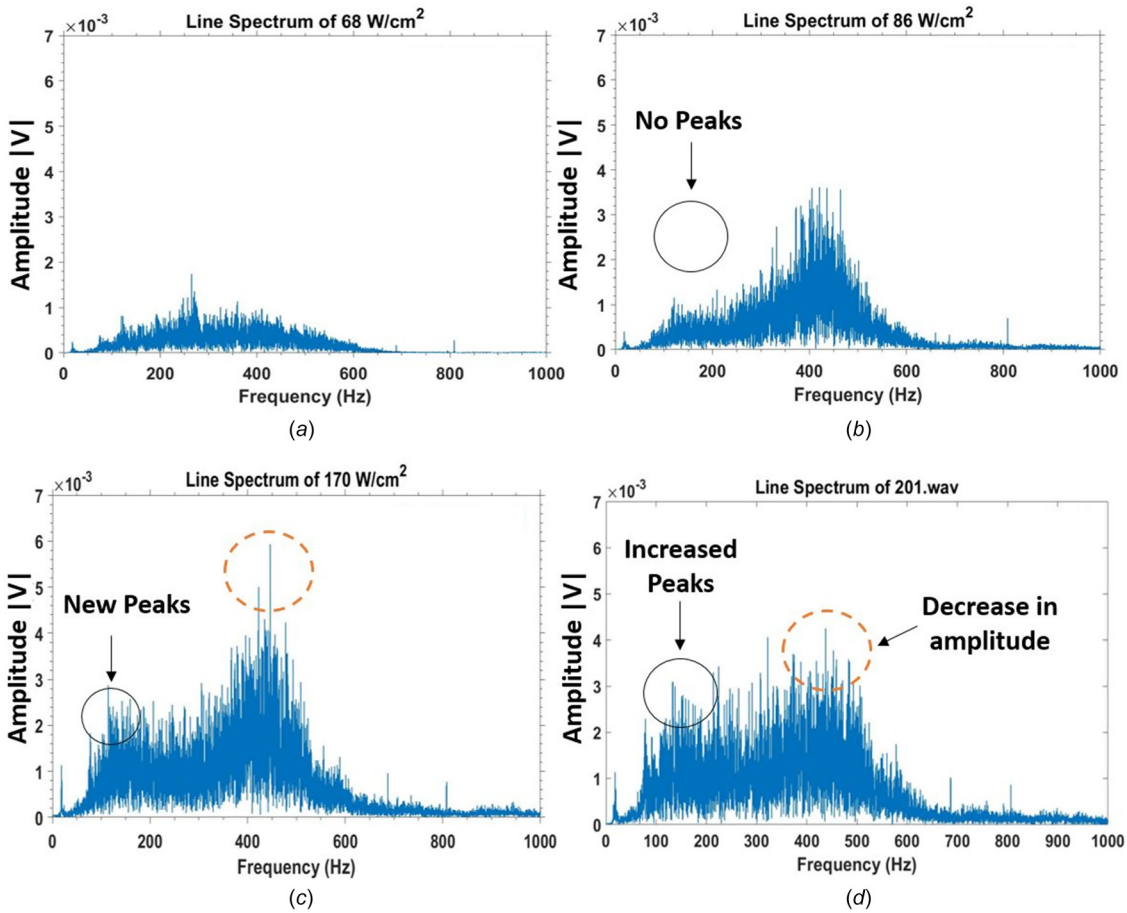


Fig. 6 Amplitude variation in frequency domain for the microporous chip at different heat fluxes (CHF = 204 W/cm^2)

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using the expression $\left(D_h^2 * D_v\right)^{\frac{1}{3}}$. The Minnaert frequencies were calculated using the following equation:

$$f = \frac{1}{2\Pi R}\sqrt{\frac{3\gamma P_O}{\rho}} \tag{6}$$

Minnaert derived Eq. (6) specifically for air bubbles of fixed mass. However, in the boiling systems, as vapor bubble grows, mass transfer takes place across the interface. This equation is thus not directly applicable for the growing boiling bubbles, but nevertheless can be used for estimating the frequency range for fully grown vapor bubbles as the mass transfer across the boundaries of fully grown vapor bubbles is reduced considerably toward the end of the bubble growth cycle. In this study, the sound is primarily produced from the oscillation and merger of fully grown vapor bubbles. So, the Minnaert equation is expected to provide a reasonably good approximation. Equivalent diameters of these vapor bubbles are measured from the high-speed images and are used in Eq. (6) to determine the frequency of sound generated by these bubbles. Figure 9 shows the corresponding high-speed images captured using a Photron high-speed camera.

Calculated Minnaert frequencies were then compared with the experimental frequencies. Tables 2 and 3 show a comparison of

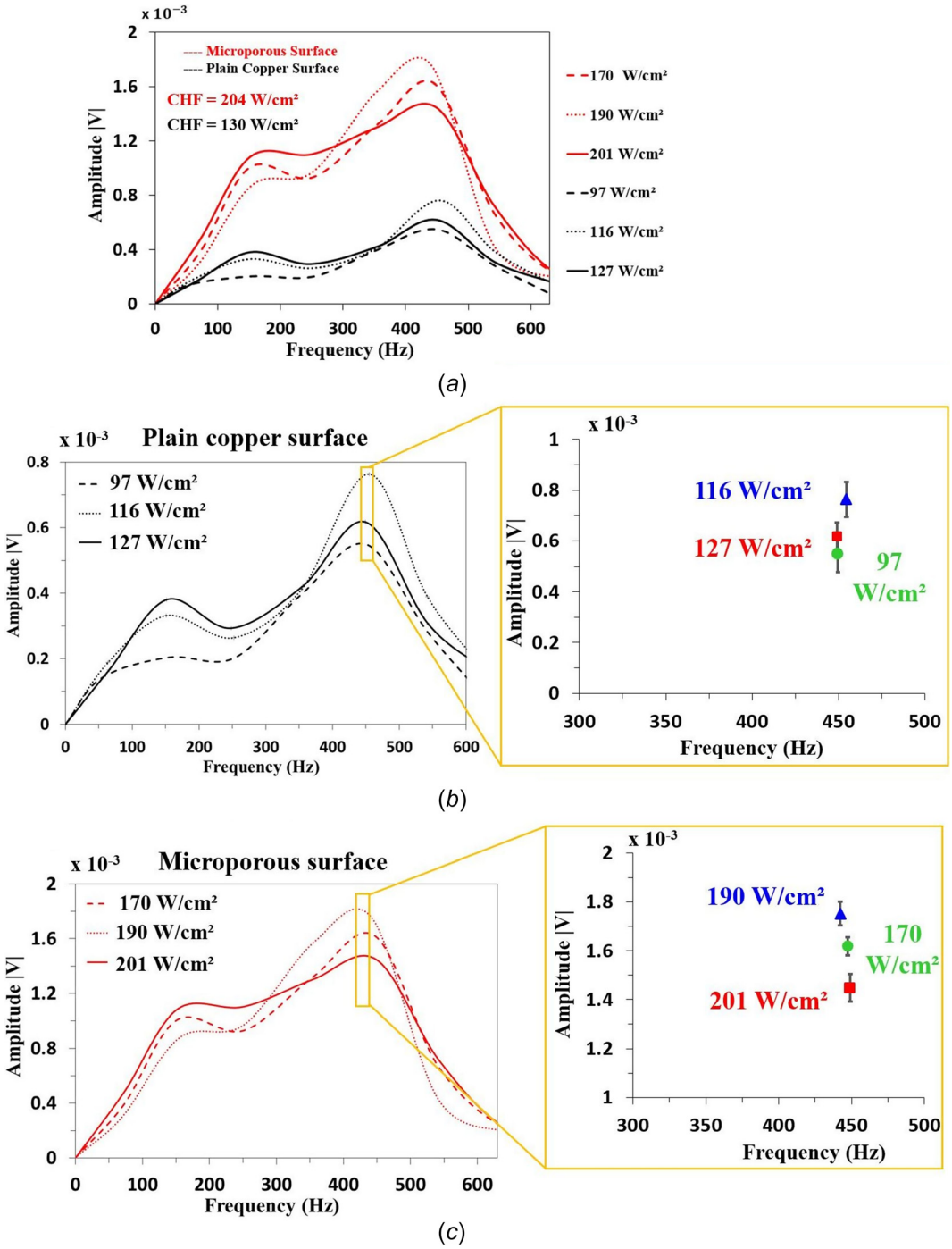


Fig. 7 (*a*) Weighted average amplitude versus frequency plot showing the variation of amplitude at high heat fluxes for a plain copper chip and a microporous chip, Amplitude error analysis for (*b*) plain copper chip, and (*c*) microporous chip at 450 Hz (the region of interest)

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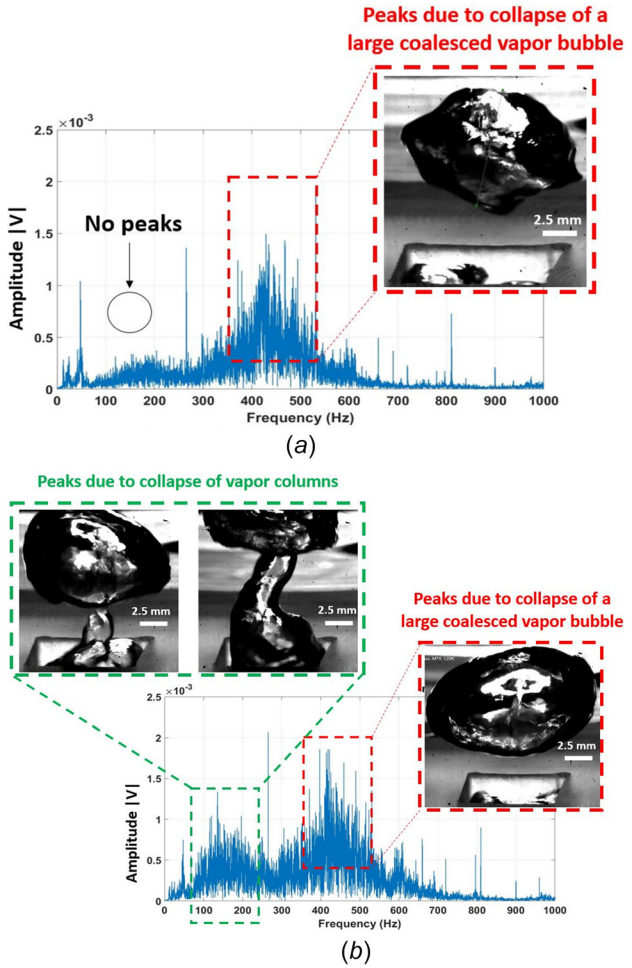


Fig. 8 High-speed images in dominant frequency regions for a plain copper chip

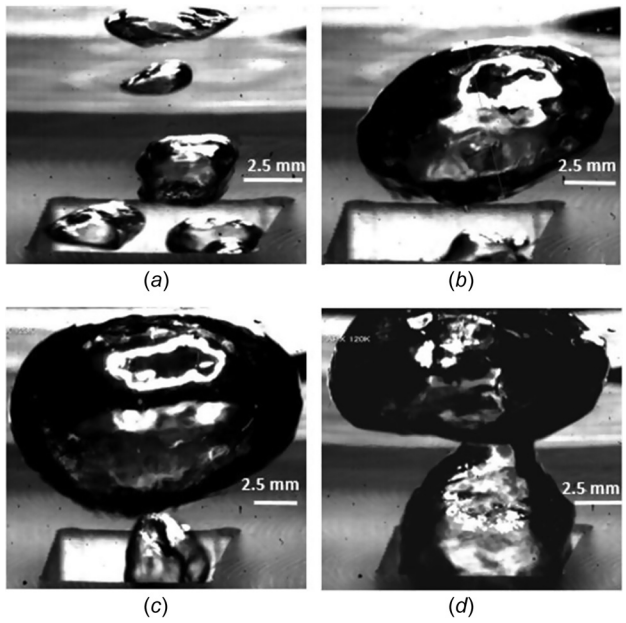


Fig. 9 High-speed images captured for a plain copper chip at (*a*) 28 W/cm^2, (*b*) 39 W/cm^2, (*c*) 96 W/cm^2, and (*d*) 126 W/cm^2

Table 2 Experimental frequency ranges at different heat fluxes

Heat flux (W/cm^2)	Experimental frequency range (Hz)
36	400–530
45	377–491
71	302–533
97	305–567
116	295–597
127	300–617

experimental frequency range and Minnaert frequency range calculated from visual data, respectively. At heat fluxes below ~50 W/cm^2, a discrepancy in experimental and calculated frequency is observed. As explained previously, the Minnaert frequency was calculated by measuring a single bubble diameter near the heater surface. From high-speed visualization it was observed that a fully developed vapor bubble grows and departs in less than ~30 ms at a heat flux around 39 W/cm^2. A similar reduction in experimental frequency with increase in number of bubbles in the bulk liquid was observed by Nikolovska et al. [23] for their air bubble experiments.

Amplitude Variations at Different Heat Fluxes. To further understand the variations in the sound signature, changes in the average amplitude at 450 Hz for a plain copper surface and 400 Hz for the microporous surface are plotted against the CHF-normalized heat flux, q''/q''_{CHF}. This plot can assist in determining the maximum heat flux at which a boiling system can safely operate after applying appropriate safety factors. Figure 10 shows the amplitude versus q''/q''_{CHF} plots for both surfaces. The thick lines in the plot represent the condition at which the heat flux is equal to the CHF value. While the dotted lines represent the last steady-state, safe operating points obtained from this study. More specifically, black dotted line indicates the upper operational limit for a plain copper surface, and red dotted line indicates the upper operational limit for the microporous surface. Since a fixed heat flux increment was applied between the two successive data points, it may be stated that a reduction of sound amplitude is indicative of the impending CHF. Such signals are however seen to be very close to the CHF; hence, fast response systems are necessary to prevent the occurrence of the CHF condition. At CHF, the entire heater surface gets covered with a vapor layer, and the sound is absent in these frequency regions at and beyond CHF .

Typically, to ensure the safety of the instruments, industrial boiling systems run only at 50–60% of the CHF condition. This safety is necessary to avoid catastrophic burnout of the systems. With the implementation of a fast response acoustic mapping, these operational ranges of the industrial boiling systems may be increased. However, further caution is warranted to validate the sound signatures generated in a specific boiling equipment while approaching so close to the CHF condition may not be advisable in systems such as nuclear reactors. Nevertheless, they can serve as a final redundant safety system. The study conducted in an open pool however shows a strong linkage between the sound amplitude and heat flux, and further study on specific equipment configuration is recommended.

Underlying Mechanism for Variation of Acoustic Trends With Heat Fluxes. Bubble activity becomes more intense during boiling with increment in heat flux (as observed from Figs. 5–7, and 10). Sound generation during the bubble formation and bubble burst is a continuous process. Initially, at lower heat fluxes, owing to lower bubble activity and fewer coalescences, amplitude intensity is lower, and time difference between successive amplitude peaks is more (as observed from Fig. 5(*a*)). With increase in heat flux as more nucleation sites become active, this time difference between two successive sound waves starts to decrease due to both increased bubble activity and bubble coalescence. Additionally, the recorded boiling sound at any particular moment is due to the combination of previous bubble burst and new bubble generation,

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Table 3 Minnaert frequency calculation from visual data

Heat flux (W/cm^2)	Measured diameter range (mm)	Minnaert frequency range (Hz)
28	5.54–6.9	966–1203
39	9.12–10.1	660–731
96	14.22–14.86	448–468
118	13.48–14.5	459–494
126	12.58–14.6	456–530

and the resultant sound is due to constructive interference of the consecutive bubbles. Based on the principle of superposition of waves, this results in higher sound amplitudes at higher heat fluxes.

In case of microporous surface, owing to the availability of additional nucleation sites for boiling, the resultant sound intensity is very high and results from constructive interference of multiple vapor bubbles nucleating from additional pores. Accordingly, higher sound amplitude for the microporous surface is noticed from Figs. 7 and 10. However, the amplitude trends for both plain and microporous surfaces are similar in a sense that the amplitude intensity increases continuously with increment in heat flux, and just before CHF, a sudden drop of amplitude is observed in the frequency range of 400-500 Hz which can be used as an indicative parameter for approaching CHF condition. By implementing faster response acoustic mapping tools, this amplitude drop can be carefully monitored, and the operational limits for the boiling systems can be increased.

Pool Boiling Curve and Corresponding Sound Signatures. Figures 11(*a*) and 11(*b*) show the pool boiling curves for a plain copper surface and a microporous surface with sound signatures at different heat fluxes. To observe the corresponding sound signatures at various heat fluxes, smoothening of the amplitude in frequency–amplitude plot was performed using weighted average formula shown in Eq. (7)

$$\bar{A} = \frac{\sum_{i=1}^{n} A_i f_i}{\sum_{i=1}^{n} f_i} \tag{7}$$

where

$$f = \frac{\sum_{i=1}^{n} A_i f_i}{\sum_{i=1}^{n} A_i}$$

For averaging, a window of 100 Hz each was considered, and average amplitude was calculated for each window. Finally, the smoothened plot for amplitude with respect to frequency was calculated for a particular heat flux by interpolating these averaged values. This plot provides the insets of both pool boiling curve and corresponding sound signatures during pool boiling on a plain copper and a microporous surface. This plot is also helpful in understanding how the sound signatures vary at different stages of nucleate boiling, and can be further used to detect the approaching CHF condition. Figure 12 shows the high-speed images of coalesced vapor bubble patterns at different heat fluxes for the plain copper surface (CHF = 130 W/cm^2) along with corresponding sound signatures. As observed from Fig. 12, the peaks in high frequency region (400–500 Hz) are due to collapse of a large coalesced vapor bubble. And at higher heat fluxes (>100 W/cm^2), peaks in lower frequency region (100–200 Hz) also appear and are due to increased vapor column formation. Just before CHF, along with bubble coalescence, increment in vapor column/jet formation is observed. This increases the sound peak intensity in lower

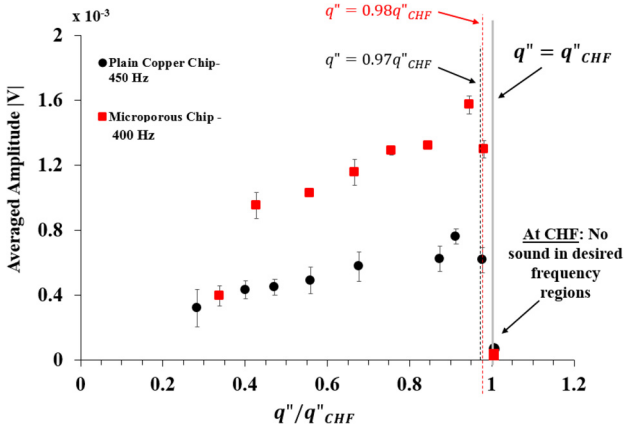


Fig. 10 Amplitude variations of a plain copper chip and a microporous chip at different heat fluxes to determine the operational range

frequency region (100–200 Hz) while reducing the peak intensity in the higher frequency region (400–500 Hz).

Limitations of this Study. This study was performed to understand the underlying mechanisms of sound generation during boiling. It also aimed at establishing theoretical and visual validation of the generated sound behavior during boiling and to understand whether the acoustic detection techniques can be implemented to detect the CHF condition during boiling and be potentially applied to enhance the operational limits of boiling equipment. A few of the limitations of this study are indicated below which should be considered before implementing acoustic techniques in real-world applications.

(1) In this study, all the experiments were performed on the pool boiling setup shown in Fig. 1. So, the variation of acoustic parameters with other test setup geometries and materials were not studied. However, preliminary tests indicated that the variation in the microphone location resulted only in a change of the amplitude, but the signature pattern remained unaltered.
(2) Distilled water was used at saturation temperature (100 °C) at an atmospheric pressure for all the experiments. The variation of acoustic parameters with a change in the liquid properties and liquid temperature is not studied, and the variation in sound parameters such as amplitude and frequency with change in pressure is also not studied. Further, the effect is not verified with different working fluids such as refrigerants, dielectric fluids, liquid mixtures and petroleum products.
(3) Microphone position in this study is kept fixed for all the tests. The change in microphone position and its effect on amplitude and frequency is not studied in this work. However, as stated in comment 1, the location of the microphone did not affect the signature pattern but only lowered the amplitude intensity.
(4) As all the tests were performed in the same lab, the background noise was constant for all the tests, which was later reduced during noise reduction. To ensure a minimum background noise, all the tests were performed during the weekends and with minimum light sources turned on (to reduce noise from fluorescent light ballast). However, to get high quality data, the tests should be performed in soundproof labs.
(5) The sensitivity of the microphone sensors used in this study is limited. So, the sound generated from bubble nucleation and bubble oscillation in the liquid could not be recorded. Since the bubble coalescence and departure were the main

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sources of sound in this study, this limitation is not expected to alter the conclusions.

(6) As microphones during all the tests are placed outside the setup, boiling sound travels through the reservoir glass and air before reaching the microphone. Reflection of sound from the glass is not analyzed, and hence, the use of hydrophone for future studies is recommended to eliminate these parameters. It is noted that this configuration of external placement of microphones is of practical interest since monitoring may be desirable from external locations in equipment such as heat exchangers and nuclear reactors.

(7) Even though a good quality microphones (frequency range: 10 Hz–10,000 Hz) and amplifier (frequency range: 20 Hz–22,000 Hz) were used in this study, higher quality

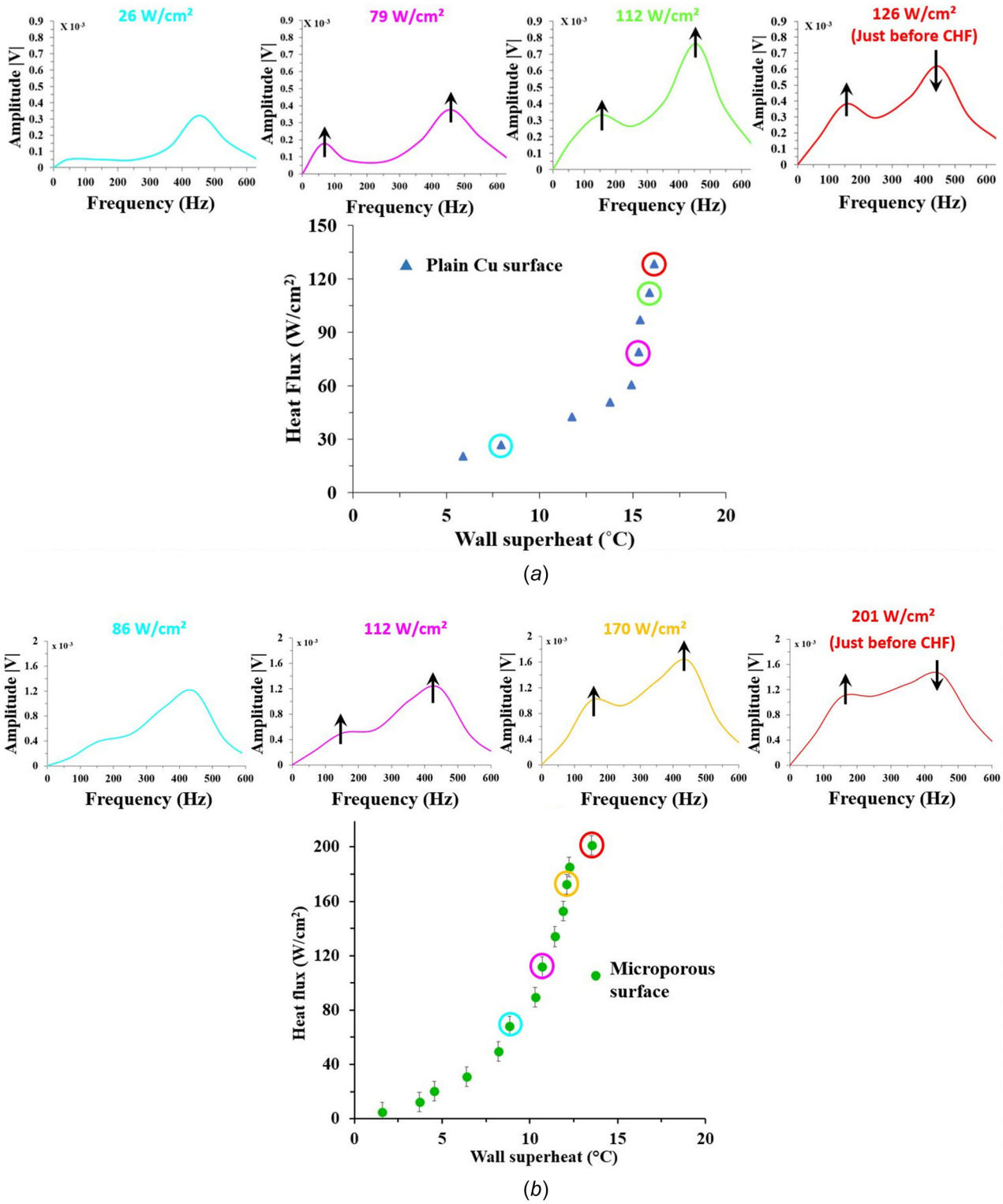


Fig. 11 Pool boiling curve and corresponding sound signatures at different heat fluxes for (*a*) a plain copper surface and (*b*) a microporous surface

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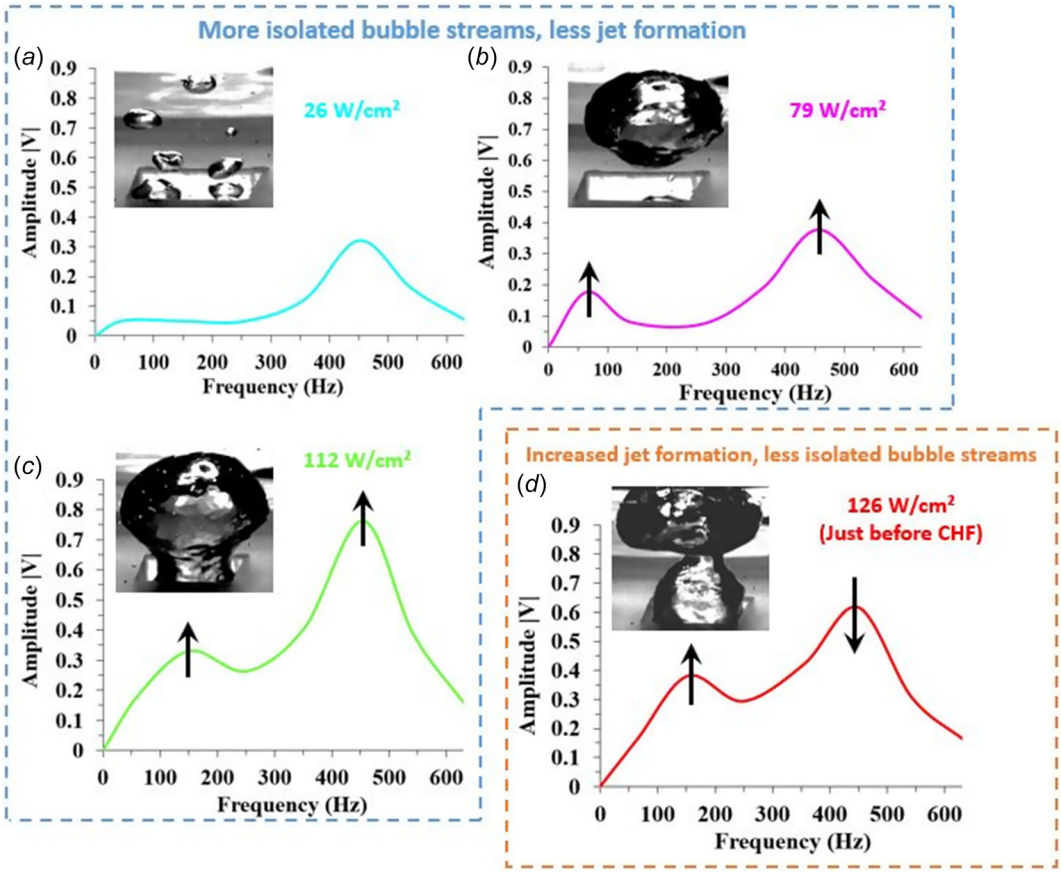


Fig. 12 High-speed bubble images and corresponding sound signatures at different heat fluxes for a plain copper surface

microphones and amplifiers covering wide frequency range with high fidelity are recommended for capturing the boiling sound data.

(8) The natural frequency of the pool boiling setup was not determined. It will be helpful to determine this frequency to ascertain that this does not interfere with the boiling sound data.

(9) Extending this work to flow boiling, since the heat fluxes are higher than pool boiling, it becomes more important to safeguard such systems. A detailed analysis needs to be performed to identify additional sources responsible for sound generation in flow boiling setups, such as pumps, throttle valves, and bends. In this study, various background noise sources were identified, and the background noise was attenuated from the recorded signal. Resultant sound after sound processing consisted of dominating boiling sound which was used for analysis. Similarly, various noise generation sources need to be identified in case of flow boiling and appropriate noise cancelation techniques can be applied to obtain clear signals for analysis.

Conclusions

Acoustic signatures during pool boiling are studied in this work. Pool boiling experiments were performed on a plain copper and a microporous enhanced surfaces. Sound data are recorded at increasing heat fluxes up to the CHF. Variation of acoustic data at each heat flux is studied and compared with bubble dynamics. The following conclusions are drawn from this study:

- The sound intensity continuously increases with an increase in heat flux due to larger coalesced bubble diameter on both plain and microporous surfaces. The rapid bubble nucleation and coalescence results in larger bubble size at high heat fluxes. A significantly higher sound amplitude was recorded in the 400–500 Hz frequency range throughout the saturated nucleate pool boiling. In addition to amplitude peaks in 400–500 Hz region at higher heat fluxes ($\sim$>100 W/cm^2), amplitude peaks in the lower frequency region (100–200 Hz) were also observed.
- With an increase in heat flux, continuous increment in amplitude was observed in both low and high frequency regions (100–200 Hz and 400–500 Hz). However, just before CHF, a sudden amplitude drop in high frequency region was observed for both plain and microporous surfaces, while the amplitude in low frequency region continued to rise. This combination of sudden drop and increment of amplitude in these two regions can be used as a tool to predict the approaching CHF condition.
- This is the first study, which correlates experimental sound frequencies during boiling with the calculated Minnaert frequencies.
- High-speed images indicate that sound generated in high frequency region is due to collapse of large coalesced vapor bubbles, while the sound in low frequency region is generated from the collapse of vapor columns.
- Pool boiling curve with insets of bubble patterns from high-speed images and frequency amplitude plots in different regions is plotted for the plain copper surface for better understanding of the boiling curve and the corresponding sound pattern.

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Technology, Rochester, NY. The authors are also thankful to Dr. Rui Liu, Assistant Professor in the Mechanical Engineering Department for his assistance in this work.

Nomenclature

CHF = critical heat flux, W/m^2
FFT = fast Fourier transform
HTC = heat transfer coefficient, W/m^2 °C
k_{Cu} = thermal conductivity of copper, W/m-K
q'' = heat flux, W/m^2
T_{sat} = saturation temperature, °C
T_{wall} = wall temperature, °C
x = distance, mm
ΔT_{sat} = wall superheat, °C

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